

# Wai Yan Min Ko Ko

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## EDUCATION

### NANYANG TECHNOLOGICAL UNIVERSITY (NTU)

Bachelor of Engineering (Hons) in Computer Engineering with Minor in Entrepreneurship

SINGAPORE

Expected Graduation: June 2028

### SINGAPORE POLYTECHNIC (SP)

Diploma with Merit in Mechatronics and Robotics (GPA: 3.90)

SINGAPORE

April 2021 – March 2024

Director's Honour Roll

Secretary & Community Program Organizer of the SP International Students' Club (SPISC)

Obtained a Professional Certificate in Computer Science for Artificial Intelligence (Harvard)

### Bukit Panjang Government High School

Singapore-Cambridge GCE O-Level Graduate

SINGAPORE

January 2019 – November 2020

MOE ASEAN Scholar (Secondary 3 – Secondary 4)

## SKILLS

### Software and Programming

- C++ & C
- Python
- HTML & CSS
- JAVA
- React
- ROS
- Linux
- Inventor
- Mastercam
- AI App Dev.
- Canva
- Prusa
- AutoCAD
- Figma

### Technical Skills

- Raspberry Pi
- Arduino
- Project Management
- Programmable Logic Controller (PLC)
- KUKA (Cobot)
- Networking
- 3D Printing
- Autonomous Mobile Robot Deployment

### Core Skills

- Problem Solving
- Adaptability
- Self-Management
- Collaboration
- Team Management

## PROJECT EXPERIENCE

### Full-stack Internship Placement Management System (Web Development)

NTU

#### Project Head & Programmer

September 2025 – November 2025

- Developed a full-stack, role-based Internship Placement Management System using Java (Spring Boot) and React, implementing secure RESTful APIs, advanced reporting, and multi-level approval workflows for students, company representatives, and staff.
- Engineered robust CSV-based data persistence (project requirements) with thread-safe operations, enforced complex business rules (eligibility, application limits, placement logic), and delivered a responsive Material-UI frontend for real-time, user-friendly internship management.

### Generative AI Lecture Companion Webapp (AI Project Development)

NTU

#### Project Head & Programmer

March 2025

- Developed a full-stack AI Lecture Companion Webapp using React for the final product and Streamlit for the prototype, enabling students to transcribe, analyze, and interact with lecture materials through audio processing and Retrieval-Augmented Generation (RAG) techniques.
- Integrated open-source AI models (Faster-Whisper for transcription and Llama3-70B via Groq API) to deliver intelligent Q&A, Key Concepts, Summarization, and Notes Taking features, enhancing the learning experience with seamless user interfaces and robust backend services.

### Automated Guided Vehicle (Arduino & Fabrication)

SINGAPORE POLYTECHNIC

#### Project Head & Programmer, Awarded Top 6 Innovative Designs

October 2022 – February 2023

- Designed, fabricated, and programmed an Automated Guided Vehicle (AGV) with a unique functional gripper that efficiently performed pick-and-place and line-tracking tasks, to guide the vehicle across a map while transporting a ball.
- Demonstrated expertise in utilizing technologies, such as an Arduino Board and diverse output shields, to power and control the Automated Guided Vehicle (AGV). Designed various AGV components using Inventor, while programming the fabrication process through Mastercam, and subsequently programmed the AGV with C++.

### Personal IOT Assistant Device (Raspberry Pi & Web Development)

SINGAPORE POLYTECHNIC

#### Project Head & Programmer

October 2022 – February 2023

- Steered a team of 2 in developing a personal IOT assistant device using a Raspberry Pi board that emulated a smart home environment.
- Collaborated with team members to develop a functional Python program and web application using HTML and CSS, implementing best practices for coding and design, and troubleshooting technical issues to ensure efficient and maintainable code.

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## PROFESSIONAL WORK EXPERIENCE

**BOTSYNC Pte. Ltd.**

**SINGAPORE**

*Mechatronics & Robotics Intern (Operations, Deployment & R&D)*

*September 2023 – February 2024*

(Key Skills: ROS, Linux, AMR Deployment, Prototyping, Client & Partner Collaboration, Project Management)

- Developed comprehensive testing of cutting-edge algorithms developed for robot navigation and workflow enhancement while collaborating proactively with the software team to implement refinements, ensuring optimal performance.
- Spearheaded compelling robot demonstrations tailored to clients, effectively showcasing the company AMRs' capabilities. Driven by a commitment to results, these demonstrations contributed significantly to increased sales and deeper client engagement.
- Directed and coordinated the seamless deployment of the AMRs at CATERPILLAR Logistics Warehouse & KEMIN Industries (Asia) Pte Ltd, ensuring the automation of their work processes.
- Provided dedicated service and support for AMRs at Bolloré Logistics (SG), Sanmina Corporation (SG), Wong Fong Engineering Works (SG), Mahindra Logistics (India), and Adyawinsa Group (Indonesia), ensuring seamless operation and client satisfaction.

**Hwa Chong Institution Boarding School (HCIBS)**

**SINGAPORE**

*Head of the Office-Student Assistant Team (Estate & IT Department) (Parttime)*

*April 2021 – Present*

(Key Skills: Web Automation Scripting, Web Design, Excel Automation, Mentorship & Training, Team Management)

- Currently lead and mentor a high-performing team of 10 NUS students, leveraging experience gained from training a diverse group of 19 student/intern staff (NUS, SMU, ITE) on essential facilities and operational procedures to ensure consistent excellence in office support.
- Streamlined the procedure for the web developer to build and launch the website, saving the web developer 1 week of labor by creating a sitemap plan for the Frontend and Backend process of HCIBS internal Defect and Cleaning Report System.
- Developed an automated Student Attendance Management Tool using Python, Selenium, and Pandas to streamline data extraction from Excel and automate attendance marking on the HCIBS portal, significantly reducing manual effort and improving data accuracy.

**Technology and Trust (TNT)**

**YANGON, MYANMAR**

*Parttime Computer Technician & Sales Assistant*

*January 2012 – October 2018*

(Key Skills: Problem-Solving, Decision Making, Customer Orientation, Communication)

- Assembled, fixed, and upgraded customers' computers and networks, boosting the processing speed of PCs, internet, and server connections, while ensuring mutual trust and strong business relationships.
- Proactively engaged with customers to identify and resolve pain points related to their computers, resulting in improved customer experience and increased sales.

## LEADERSHIP EXPERIENCE

**Singapore Polytechnic International Students' Club (SPISC)**

**SINGAPORE**

*Secretary & Community Program Organizer of the SP International Students' Club*

*July 2021 – March 2024*

(Key Skills: Multiple Stakeholders, Communication, Self & Team Management, Problem Solving, Creative Thinking)

- Provide a fun work environment where CCA members across 8 departments can collaborate and accomplish their objectives for all CCA events while tracking CCA matters and taking care of the members' welfare.
- Coordinated with the school administration, faculty, and staff to ensure smooth collaboration and participation in events, resulting in high turnout and engagement from students, with at least 150 student participants per event.

## COMPETITION EXPERIENCE

**Google Data Centre Outreach Hackathon**

**SINGAPORE**

*4<sup>th</sup> Place Winner - Team*

*October 2025*

(Key Skills: Linux, Networking, Server Assembly, Troubleshooting, Communication)

- Directed a six-person team for rapid physical assembly of a disassembled enterprise server (fans, power supply, storage, and network cabling), leveraging prior PC-building expertise to accelerate the hardware phase and ensuring team success in the initial critical stage.
- Resolved complex server-to-VM networking issues and debugged unfamiliar server BIOS configurations under a strict time constraint, navigating enterprise hardware interfaces and applying methodical troubleshooting steps beyond standard PC systems.